

The complete complex amplitude in gamma sum descent

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1 Original objects and the question answered

The delivered Gamma Convolution Descent note constructs the conditional expectation of the squared arithmetic amplitude. We construct the exact comparison for its full complex amplitude. The two resulting source Grams add to the original arithmetic Gram. We prove a strict positive contribution of the retained relative component for every full quartet, and give its exact map into the original minimum quotient metric. All measures below retain their masses. The variable is the original sum $u = t_1 + \cdots + t_k$, and the arithmetic variable remains $S = k/2 + iu$.

Fix $\lambda > 0$ and an integer $k \geq 1$. Retain

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad r_{\lambda,k} = r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}. \quad (\text{GP.1})$$

Thus $\int r_{\lambda,k} = c_\lambda^k$. The beta-integral proof of this identity, with its original Jacobian $t = 2v$, is retained in the cumulative reader, TG.7–TG.8, and the delivered note, (5)–(9). In particular

$$\int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-2k\lambda}, \quad |\theta| < \pi/2. \quad (\text{GP.2})$$

Let h be the original finite polynomial packet of complete orders of actual zeros of $g = 2\xi$, and include $h = 1$ when considering the analytic seed. Define the entire quotient $v_h = g/h$ and keep

$$A(t) = \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B(t) = |A(t)|^2, \quad w(t) = B(t)r_\lambda(t), \quad \mu_h = \int w(t) dt. \quad (\text{GP.3})$$

Use one of the references proved in the delivered note for which $B \leq C_h < \infty$. In particular a full quartet of degree $d \geq 4$ uses its original $\lambda = 1/4$. The seed uses $\lambda = 13/4$ and $C_1 = 1024/\sqrt{\pi}$. The statements involving decay explicitly use the quartet reference. The function A is nonzero and analytic on the real line: v_h is a nonzero entire function and reciprocal gamma is entire and nonzero on this line. Its real zeros form a discrete set. No value at a removed packet zero is assigned using an uncanceled fraction.

Put $d\nu(\mathbf{t}) = \prod_i r_\lambda(t_i) d\mathbf{t}$ and $d\nu_\Sigma(u) = r_{\lambda,k}(u) du$. In $\mathcal{H} = L^2(\nu)$ and $\mathcal{H}_\Sigma = L^2(\nu_\Sigma)$ write

$$Uf(\mathbf{t}) = f(\sum_i t_i), \quad C = U^*, \quad P = UC, \quad a(\mathbf{t}) = \prod_i A(t_i), \quad (\text{GP.4})$$

$$b_k = C(|a|^2), \quad a_k = C(a), \quad d_k = b_k - |a_k|^2.$$

For $k = 1$, $C = U = I$, $a_1 = A$, $b_1 = B$, $d_1 = 0$. For $k \geq 2$ the conditional integral CF is precisely

$$(CF)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GP.5})$$

These are probability measures on individual fibres, derived by dividing by their stated fibre mass; neither original measure is replaced by a probability measure. Fubini proves $CU = I$ and the isometry of U . Jensen's inequality in (GP.5) proves $d_k \geq 0$ and $0 \leq b_k \leq C_h^k$. For $k \geq 2$, $b_k(u) > 0$ at every real u : the discrete zeros of each factor exclude only countably many hyperplanes in this fibre, which have zero Lebesgue measure.

2 Unitary transport of the actual conditional projection

Theorem 1 (Complete amplitude comparison). *Use the actual arithmetic spaces*

$$\mathcal{H}^h = L^2(\mathbb{R}^k, |a|^2 d\nu), \quad \mathcal{H}_\Sigma^h = L^2(\mathbb{R}, b_k d\nu_\Sigma).$$

The maps

$$T : \mathcal{H}^h \longrightarrow \mathcal{H}, \quad F \longmapsto aF, \quad T_\Sigma : \mathcal{H}_\Sigma^h \longrightarrow \mathcal{H}_\Sigma, \quad f \longmapsto \sqrt{b_k}f \quad (\text{GP.6})$$

are surjective isometries. The positive square root in T_Σ is a specified comparison map; the original amplitude in T remains complex. The actual arithmetic sum inclusion $U_h f = f(\sum t_i)$ and its adjoint C_h^{sum} satisfy

$$C_h^{\text{sum}} F = \frac{C(|a|^2 F)}{b_k}, \quad J = TU_h T_\Sigma^{-1} = M_a U M_{b_k^{-1/2}}, \quad J^* = M_{b_k^{-1/2}} C M_{\bar{a}}, \quad J^* J = I. \quad (\text{GP.7})$$

All quotients here are defined almost everywhere. In particular their operator extensions are bounded even when the individual inverse multiplier is unbounded. The transported arithmetic projection is

$$Q = TP_h T^{-1} = JJ^*, \quad QF = aU \left(\frac{C(\bar{a}F)}{b_k} \right), \quad P_h = U_h C_h^{\text{sum}}. \quad (\text{GP.8})$$

Thus P and Q are related by the displayed exact transport from the original arithmetic spaces.

Proof. The squared norms in (GP.6) agree by direct substitution. The zero set of a is a union of null hyperplanes; b_k is positive almost everywhere, including $k = 1$. Division off those null sets constructs the two inverses with their exact squared norms. Applying (GP.5) to the arithmetic fibre density gives C_h^{sum} . Pairing $M_a U M_{b_k^{-1/2}} f$ with F proves the stated adjoint. The identity $C(|a|^2 Uf) = b_k f$ then gives $J^* J = I$. Conjugating P_h gives (GP.8). These norm identities first hold for bounded functions supported on finite-measure sets on which the denominators are bounded below; truncation and the isometries extend them to the indicated complete Hilbert spaces. This also proves that no unbounded multiplier composition is being asserted on a larger domain. $\square$

Theorem 2 (Exact fibre angle and retained defect). *Define $\gamma_k = a_k / \sqrt{b_k}$ almost everywhere. Then*

$$U^* J = M_{\gamma_k}, \quad J = U M_{\gamma_k} + N, \quad U^* N = 0, \quad N^* N = M_{1-|\gamma_k|^2} = M_{d_k/b_k}. \quad (\text{GP.9})$$

For every real u at which $b_k > 0$, the two fibre projections onto 1 and $a/\sqrt{b_k}$ obey

$$\text{Tr}_u(P_u - Q_u)^2 = 2(1 - |\gamma_k(u)|^2), \quad \|[P_u, Q_u]\|_{\text{HS},u}^2 = 2|\gamma_k(u)|^2(1 - |\gamma_k(u)|^2). \quad (\text{GP.10})$$

These are finite-rank traces in the specified fibre Hilbert space, not a claim that the global decomposable operators are trace class. The complete unscaled variance also has the formula

$$d_k(u) = \frac{1}{2} \iint |a(\mathbf{t}) - a(\mathbf{s})|^2 d\nu_u(\mathbf{t}) d\nu_u(\mathbf{s}), \quad (\text{GP.11})$$

where both ν_u are the exact conditional measures in (GP.5).

Proof. Applying C to Jf gives $\gamma_k f$. Orthogonal decomposition with P gives (GP.9), and taking squared norms uses $J^*J = I$. On a fibre let $e_0 = 1$ and $v = a/\sqrt{b_k}$, so both have norm one and $\langle e_0, v \rangle = \gamma_k$, using the inner product antilinear in its first argument. If $|\gamma_k| < 1$, put $e_1 = (v - \gamma_k e_0)/\sqrt{1 - |\gamma_k|^2}$. In the displayed orthonormal comparison frame, without altering either original vector, the two projections are

$$P_u = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q_u = \begin{pmatrix} \frac{|\gamma_k|^2}{\gamma_k \sqrt{1 - |\gamma_k|^2}} & \gamma_k \sqrt{1 - |\gamma_k|^2} \\ \gamma_k \sqrt{1 - |\gamma_k|^2} & 1 - |\gamma_k|^2 \end{pmatrix}.$$

Matrix multiplication proves (GP.10); both vanish when $|\gamma_k| = 1$. Expanding the square in (GP.11), and using the unit total mass of each conditional measure, gives $C(|a|^2) - |Ca|^2$. No phase is discarded in this calculation. $\square$

3 Strictness for an actual quartet, with its exact constant

For a full quartet retain $\lambda = 1/4$, degree $d \geq 4$ and $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$. The delivered proof gives

$$|A(t)|^2 \leq \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} |t|^{6-2d} \quad (|t| \geq T_h). \quad (\text{GP.12})$$

Here is its direct calculation. In the unchanged amplitude,

$$A(t) = -\pi^{-1/4} (t^2 + 1/4) e^{-it \log \pi/2} \zeta(1/2 + it) / h(1/2 + it).$$

The continuation formula $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$ gives $|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|)$. For $|t| \geq T_h$, every retained root factor has modulus at least $|t|/2$, so $|h| \geq (|t|/2)^d$. Also $(t^2 + 1/4)^2 \leq 25t^4/16$ and $4(1 + |t|)^2 \leq 16t^2$ for $|t| \geq 1$. Multiplication proves (GP.12). Consequently $A(t) \rightarrow 0$ at both real ends.

Theorem 3 (Strict relative contribution on every sum fibre). *For every such actual quartet, every $k \geq 2$, and every real u ,*

$$d_k(u) > 0, \quad |\gamma_k(u)| < 1. \quad (\text{GP.13})$$

Thus the relative component is nonzero at every fibre. It is given exactly by (GP.9) and (GP.11); strictness does not assign a uniform positive lower bound to it.

Proof. Fix u . If $d_k(u) = 0$, (GP.11), or equality in the variance identity, says that the continuous function a is constant almost everywhere on the fibre. Its conditional density is strictly positive everywhere, so continuity makes it constant everywhere. For $k = 2$, this would make $A(t)A(u-t)$ constant in t . Its limit is zero by (GP.12) and boundedness of A , whereas some value is nonzero: avoid the two discrete sets of zeros of $A(t)$ and $A(u-t)$. For $k > 2$, choose fixed $t_2, \dots, t_{k-1}$ at nonzeros of A and vary $t_1 = t$, $t_k = u - t - \sum_{2 \leq i < k} t_i$. The identical argument makes the product both zero and nonzero, a contradiction. The assertion for γ_k follows by dividing by the strictly positive b_k already proved. This proof uses only the explicit quartet tail, not a conjecture about positions of unselected zeros. $\square$

4 Full complex coefficient transform and exact variance norm

Put $\alpha = 2\lambda$ and keep the original monic polynomials

$$\sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2}, \quad \|b_n^{(\lambda)}\|^2 = c_\lambda n! (\alpha)_n. \quad (\text{GP.14})$$

The specified gamma measure has exponential moments. Its polynomials are complete in L^2 : for a vector orthogonal to them, multiplying it by the measure gives a finite complex measure by Cauchy–Schwarz; the same inequality with exponential moments makes its Fourier transform holomorphic on a strip about the real axis. All derivatives at zero vanish. Analytic uniqueness and Fourier uniqueness make the measure, hence the vector, zero. The same proof applies to the sum measure with mass c_λ^k .

Define complex coefficients, retaining conjugation conventions,

$$\mathbf{a}_j = \frac{\int b_j^{(\lambda)}(t)A(t)r_\lambda(t)dt}{c_\lambda j!(\alpha)_j}, \quad \mathfrak{A}(z) = \sum_{j \geq 0} (\alpha)_j \mathbf{a}_j z^j, \quad e_{k,n} = [z^n] \mathfrak{A}(z)^k. \quad (\text{GP.15})$$

The square-integrable vector A has these orthogonal coefficients. The series in (GP.15) may be used coefficientwise; no interchange of an infinite product with an integral is required below.

Theorem 4 (Complex convolution and its exact lost norm). *With all original masses,*

$$a_k(u) = \sum_{n \geq 0} \frac{e_{k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u) \quad \text{in } L^2(r_{\lambda,k} du), \quad (\text{GP.16})$$

$$\int d_k r_{\lambda,k} = \mu_h^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |e_{k,n}|^2.$$

For every polynomial F in the original S coordinate and every real $|\theta| < \pi/2$, the exact weighted version is

$$\begin{aligned} \|a U(e^{\theta u/2} F(k/2 + iu))\|^2 &= \int e^{\theta u} |F(k/2 + iu)|^2 |a_k(u)|^2 r_{\lambda,k}(u) du \\ &+ \int e^{\theta u} |F(k/2 + iu)|^2 d_k(u) r_{\lambda,k}(u) du. \end{aligned} \quad (\text{GP.17})$$

Proof. Multiplication of the finite-degree coefficients of the generating functions gives

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i).$$

Pairing this finite identity with a gives $c_\lambda^k n! e_{k,n}$. Division by the original sum norm $c_\lambda^k n! (k\alpha)_n$, followed by completeness, gives the first identity. Pythagoras for $P = UC$ gives the second since $\|a\|^2 = \mu_h^k$. Finally conditional expectation commutes with multiplication by a function of the sum. Apply the same orthogonal decomposition to $a U(e^{\theta u/2} F)$ to obtain (GP.17). Boundedness by C_h^k , (GP.2), and polynomial growth prove all integrals finite, including any fixed number of theta derivatives. $\square$

5 The exact induced change of the original finite quotient metric

Fix the actual annihilator $\chi = \chi_{h,k}$, degree $q > 0$, and $N \geq q - 1$. Use original polynomial spaces $E = \mathcal{P}_N(S)$, $E_\partial = \mathcal{P}_{N-q}(S)$, where a negative degree means the zero space, and $C_\chi = \mathbb{C}[S]/(\chi)$. Let $B_\chi : E_\partial \rightarrow E$ be multiplication by χ and $\pi : E \rightarrow C_\chi$ the quotient. Use the original monomial bases, or any explicitly transported fixed bases, in every matrix below. For a coefficient column x denote its polynomial by F_x . Define three source matrices by

$$\begin{aligned} x^* M_h x &= \int e^{\theta u} |F_x(k/2 + iu)|^2 b_k r_{\lambda,k} du, \\ x^* M_{\text{coh}} x &= \int e^{\theta u} |F_x(k/2 + iu)|^2 |a_k|^2 r_{\lambda,k} du, \\ x^* M_{\text{rel}} x &= \int e^{\theta u} |F_x(k/2 + iu)|^2 d_k r_{\lambda,k} du. \end{aligned} \quad (\text{GP.18})$$

These matrices are elements of $\text{Herm}(E)$, equivalently linear maps $E \rightarrow E^\vee$ with $E^\vee$ the conjugate dual under $x \mapsto [y \mapsto y^* M x]$. The associated linear observation maps are $F \mapsto aU(e^{\theta u/2} F)$, $F \mapsto U(e^{\theta u/2} a_k F)$, and $F \mapsto (I - P)aU(e^{\theta u/2} F)$. The Gram operation itself has type $\text{Hom}_{\mathbb{C}}(E, \mathcal{H}) \rightarrow \text{Herm}(E)$, $R \mapsto R^* R$. Thus

$$M_h = M_{\text{coh}} + M_{\text{rel}}. \quad (\text{GP.19})$$

The original arithmetic matrix M_h is unchanged.

Lemma 5. *M_{coh} is positive definite for every packet/reference admitted above, every $k \geq 1$, every finite N , and real $|\theta| < \pi/2$. For the quartets of (GP.12) and $k \geq 2$, M_{rel} is also positive definite.*

Proof. Put $f = Ar_\lambda$. It is a nonzero L^1 function, since A is bounded and nonzero almost everywhere. The numerator of a_k is f^{*k} . If this convolution were zero almost everywhere, its Fourier transform $(\hat{f})^k$ would vanish everywhere, whence $\hat{f} = 0$ and $f = 0$ by Fourier uniqueness, a contradiction. Consequently $|a_k|^2 r_{\lambda,k}$ is positive on a set of positive Lebesgue measure. A nonzero polynomial on the indicated affine line has only finitely many real zeros, so its squared norm is strictly positive. This proves the first assertion. The second follows from $d_k(u) > 0$ for every u , proved in (GP.13). $\square$

Let $R_h, R_{\text{coh}} : C_\chi \rightarrow E$ be the least-norm sections for M_h, M_{coh} , and write $G_h = R_h^* M_h R_h$, $G_{\text{coh}} = R_{\text{coh}}^* M_{\text{coh}} R_{\text{coh}}$. They exist uniquely: for any section R_0 , subtract $B_\chi(B_\chi^* M B_\chi)^{-1} B_\chi^* M R_0$; the boundary Gram is invertible because B_χ is injective and M positive definite. An empty boundary contributes the zero map.

Theorem 6 (Exact two positive terms in the quotient change). *In these original coordinates put*

$$K = (B_\chi^* M_h B_\chi)^{-1} B_\chi^* M_{\text{rel}} R_{\text{coh}}. \quad (\text{GP.20})$$

It is a map $C_\chi \rightarrow E_\partial$, or the unique zero map when the boundary is empty. Then

$$R_h = R_{\text{coh}} - B_\chi K, \quad \boxed{G_h = G_{\text{coh}} + K^* B_\chi^* M_{\text{coh}} B_\chi K + R_h^* M_{\text{rel}} R_h}. \quad (\text{GP.21})$$

For an actual full quartet, $k \geq 2$, and $q > 0$, this gives $G_h - G_{\text{coh}} > 0$ for every admitted N and theta. More precisely, with the actual positive matrix

$$E_q = G_{\text{coh}}^{-1/2} (K^* B_\chi^* M_{\text{coh}} B_\chi K + R_h^* M_{\text{rel}} R_h) G_{\text{coh}}^{-1/2}, \quad (\text{GP.22})$$

the original determinants satisfy

$$\det G_h = \det G_{\text{coh}} \det(I + E_q). \quad (\text{GP.23})$$

These comparison conjugations do not replace either quotient metric.

Proof. $B_\chi^* M_{\text{coh}} R_{\text{coh}} = 0$, so the formula for the M_h section using R_{coh} as initial section gives (GP.20) and the first identity. Expand $R_h^* M_{\text{coh}} R_h$ about R_{coh} . The two cross terms vanish by the stated orthogonality. Add $R_h^* M_{\text{rel}} R_h$ using (GP.19); this proves (GP.21) with both terms retained. R_h is injective because $\pi R_h = I$. Thus strict positivity of M_{rel} makes its displayed pullback positive definite on the nonzero quotient. Conjugation by the actual positive square root of G_{coh} proves (GP.23). For $N = q - 1$ the first positive term is zero and the second remains as stated. For $q = 0$ the original quotient is zero: both section maps are zero, and the empty determinant is one. The nonzero analytic source does not supply an extra quotient dimension. $\square$

6 Support, the original unit, and the domain of arithmetic observation

The original cyclic source is $\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}$. In its Mellin realization its full image is

$$\prod_i \frac{\Gamma(\lambda + it_i/2)}{\sqrt{2\pi}} a(\mathbf{t}) P(k/2 + i \sum_i t_i). \quad (\text{GP.24})$$

Multiplication by the displayed gamma factors is an isometry $\mathcal{H} \rightarrow L^2(dt)$. Equations (GP.9) and (GP.17) split the vector before this multiplication and reconstruct it by addition. On the image graph

$$\mathcal{G}_N = \{(U(a_k F), (I - P)aUF) : F \in \mathcal{P}_N(S)\}$$

addition $\text{add}_N : \mathcal{G}_N \rightarrow aU(\mathcal{P}_N)$ is injective, since the squared norm of the reconstructed vector is the strictly positive polynomial norm M_h . It is onto the displayed image by construction. Denote the multiplication in (GP.24) by $M_\Gamma : \mathcal{H} \rightarrow L^2(dt)$ and the original unitary Mellin map by $\mathcal{M}_{\text{unit}}$. The graph observation is exactly $J^{(k)} \mathcal{M}_{\text{unit}}^{-1} M_\Gamma \text{add}_N$, and likewise with $q^{(k)}$. These are well-defined linear maps because the reconstructed image is precisely the original admitted Mellin source. Their values on the graph of P are exactly

$$J^{(k)} \mathcal{V}P = \eta_{h,k} \pi P, \quad q^{(k)} \mathcal{V}P = \sigma_h^{\otimes k} \eta_{h,k} \pi P, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_k) 1. \quad (\text{GP.25})$$

This specifies the domain of the observation; no arithmetic jet is assigned to an arbitrary independent measurable relative vector. The graph in this display uses theta zero. At a nonzero admitted theta, multiply both graph components by the same $e^{\theta u/2}$; its inverse multiplies by $e^{-\theta u/2}$ before the original inverse Mellin map. Both maps are isometries between the declared tilted and untilted spaces.

Here is the full cochain map for the relation. In the original tensor algebra

$$\mathbb{C}[s_1, \dots, s_k] / (h(s_1), \dots, h(s_k)),$$

the cyclic annihilator satisfies $\chi(s_1 + \cdots + s_k) = 0$. Successive monic division in the fixed order $s_1, \dots, s_k$ gives unique quotients at each division step and polynomials Q_i with

$$\chi(s_1 + \cdots + s_k) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k).$$

The final remainder is zero because the monomials of degree less than $\deg h$ in each variable are a basis of that tensor algebra. In $[V \xrightarrow{\Theta} \mathcal{B}]^{\otimes k}$ put $D = -x\partial_x$ and $\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. It is even, $\phi_0(0) = 0$ and $\int_{\mathbb{R}} \phi_0 = 0$: the two even Gaussian integrals are $4\pi^2 \cdot 3/(4\pi^2)$ and $6\pi \cdot 1/(2\pi)$, both equal to three. For $\Re s > 1$, termwise Mellin integration of $\Theta\phi_0(x) = \sum_{n \neq 0} \phi_0(nx)$ yields $2\zeta(s) \mathcal{M}\phi_0(s)$. Direct integration with $y = \pi x^2$ gives

$$\mathcal{M}\phi_0(s) = \pi^{-s/2} (2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2).$$

Thus $\mathcal{M}(\Theta\phi_0) = g$ there and by continuation on the original Mellin domain. The original inverse Mellin definition of F_h then gives $h(D)F_h = \Theta\phi_0$. The precise primitive is

$$\mathcal{P}_\chi F = \sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) F(D_1 + \cdots + D_k) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{GP.26})$$

Euler differentiation preserves the even Schwartz source and its two conditions: its value at zero vanishes and its integral is the original integral by integration by parts. It also preserves

rapid decay at both multiplicative ends and commutes with Θ . Every term therefore lies in the asserted cochain domain. The preceding $i - 1$ degree-one factors give the tensor differential sign $(-1)^{i-1}$, cancelling the written sign. Hence, with no sign omitted,

$$d\mathcal{P}_\chi F = \mathcal{V}(\chi F), \quad d[-\mathcal{P}_\chi K] = \mathcal{V}(R_h - R_{\text{coh}}). \quad (\text{GP.27})$$

This proves the cochain correction map on C_χ in (GP.21), including its zero value when the boundary is empty.

Every linear map on this graph and every original source or quotient has its original fixed-support lift: $(\ell, x) \mapsto (\ell, Lx)$, and external absence maps to external absence. The maps preserve addition and scalar action because the underlying maps are linear. For an actual relation the quotient value is $(\ell, 0)$, its supported zero, with the source and the two graph components still reconstructible. The norm has type $\mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$ and is positively homogeneous; its square is quadratic. Trace has type $\text{End}_{\mathbb{C}}(E) \rightarrow \mathbb{C}$ and is linear; its restriction to Hermitian endomorphisms is real-linear. The Gram map and squared modulus are quadratic on the underlying real vector spaces, and trace composed with a Gram map is quadratic. The exact linear observation maps producing the source Grams and the quotient values are (GP.18), (GP.21), and (GP.24)–(GP.25).

7 Exact polynomial calibration with the original gamma mass

The following auxiliary calibration tests the complete complex-phase identities by rational arithmetic. It is a declared polynomial amplitude, with all fixed moments finite in the original gamma measure. It does not invoke the bounded packet estimate (GP.12). Take $\lambda = 1$, $k = 2$, $c_1 = 1/2$, $c_1^2 = 1/4$, and $A(t) = 1 + it/2$. The original sum coordinate is $S = 1 + iu$. Direct conditional polynomial calculation gives

$$\begin{aligned} a_2(u) &= \frac{6}{5} + \frac{iu}{2} - \frac{u^2}{20}, \\ b_2(u) &= \frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120}, \\ d_2(u) &= \frac{18}{175} + \frac{13u^2}{1400} + \frac{u^4}{5600} > 0. \end{aligned} \quad (\text{GP.28})$$

Here are enough exact polynomial identities to prove the calculation. For the one-factor parameter $\alpha = 2$, $b_2^{(1)}(t) = t^2 - 2$. For the sum parameter 4, the second and fourth monic polynomials are $u^2 - 4$ and $u^4 - 32u^2 + 72$. The conditional addition formula yields

$$\begin{aligned} C(t_1 t_2) &= (u^2 - 4)/5, \\ C(t_1^2) &= C(t_2^2) = 3(u^2 - 4)/10 + 2, \\ C(t_1^2 t_2^2) &= \frac{3}{70}(u^4 - 32u^2 + 72) + \frac{6}{5}(u^2 - 4) + 4. \end{aligned}$$

Indeed the respective coefficients are $2^2/(4)_2$, $(2)_2/(4)_2$, and $(2)_2^2/(4)_4$. Expanding $(1 + it_1/2)(1 + it_2/2)$ and $(1 + t_1^2/4)(1 + t_2^2/4)$ using these identities gives the first two lines of (GP.28). Subtracting the squared modulus gives its last line. This is the complete complex subtraction, including its imaginary linear coefficient.

For the auxiliary annihilator $\chi(S) = (S - 1)^2 - 2$, use the original S monomials at $N = 1$, where the boundary is empty. The sum gamma moments of orders zero, two, four, six are

respectively $1/4, 1, 14, 376$, obtained by differentiating $\frac{1}{4} \cos^{-4} \theta$. Substitution in (GP.18) gives

$$G_h = \begin{pmatrix} 9/16 & 9/16 \\ 9/16 & 81/16 \end{pmatrix}, \quad G_{\text{coh}} = \begin{pmatrix} 21/40 & 21/40 \\ 21/40 & 189/40 \end{pmatrix},$$

$$G_h - G_{\text{coh}} = \begin{pmatrix} 3/80 & 3/80 \\ 3/80 & 27/80 \end{pmatrix}, \quad (\text{GP.29})$$

$$\det G_h = 81/32, \quad \det G_{\text{coh}} = 441/200, \quad \det(G_h - G_{\text{coh}}) = 9/800.$$

For example $\int b_2 r_{1,2} = 9/16$ and $\int u^2 b_2 r_{1,2} = 9/2$, while the corresponding coherent moments are $21/40$ and $21/5$. Odd moments vanish, so the original off-diagonal S entries equal the masses. These calculations prove every entry above. They exhibit the strictly positive contribution in the unchanged quotient coordinates.

The accompanying checker additionally derives moments from the Laplace function, solves the independent product-moment equations, and checks all moments through degree eight and the section/quotient identities through $N = 4$. It uses a separate two-atom conditional fibre of original masses two and three to test the complex adjoint and both projection traces. Its mathematical controls change a conjugate transpose, omit an original mass factor, or omit the relative quotient term. Those finite checks supplement the proofs above and do not estimate an actual growing-degree zeta quotient.

8 Provenance and consequence

The source constants, packet, conditional density, and coefficient addition identity come from the complete delivered Gamma Convolution Descent note, equations (2), (5)–(19), (24)–(27), and (32). The original unit and theta relation primitive come from the retained cyclic source construction and TVB in the cumulative reader. The classical generating function is NIST DLMF 18.23.7, with $b_n^{(\lambda)}(t) = n! P_n^{(\lambda)}(t/2; \pi/2)$. The exact specialization and norms are proved in TG.7–TG.14. The bounded amplitude is used here to prove the full phase transport, fibre projection formulas, strict quartet variance, complex coefficient transform, and two-term positive quotient identity. These equations provide additional exact data for the same Toda quotient. They do not give a growing-degree estimate for its consecutive volume ratio. That ratio remains the actual determinant expression of the retained metric, now with the explicit contribution (GP.21)–(GP.23).